

BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE421 / EEE465: Computer Networks — Final Examination

Semester: Summer 2025 — Set: B — Full Marks: 70 — Duration: 2 Hours

COMPREHENSIVE SOLUTION AND COMPLETE ANSWER KEY

SECTION A: MANDATORY QUESTIONS (40 MARKS)

Question 1: Subnet Identification, Directed Broadcast, & VLSM [CO3] 2 + 2 + 10 + 2 = 16 Marks

Problem Context: The number of hosts given in the topology only includes PCs, printers, and servers. Sheldon and Leonard are both on R1-LAN. Their IP addresses are 192.0.50.69 and 192.127.255.253 (2nd last usable IP address of the network), respectively. Penny is on this network and uses the first usable IP address.

I. Calculate the subnet mask of R1-LAN based on the above scenario. [2 Marks]

Solution:

- Leonard's IP is the 2nd last usable IP: 192.127.255.253.
- Therefore, the last usable IP is 192.127.255.254, and the broadcast address is 192.127.255.255.
- The 2nd octet of the broadcast address is 127 (01111111₂).
- The network block spans from 192.0.0.0 to 192.127.255.255, which encompasses 128 blocks in the 2nd octet ($2^7 = 128$).
- Prefix length: $8 + (8 - 7) = /9$.
- Dotted-decimal subnet mask: **255.128.0.0**.

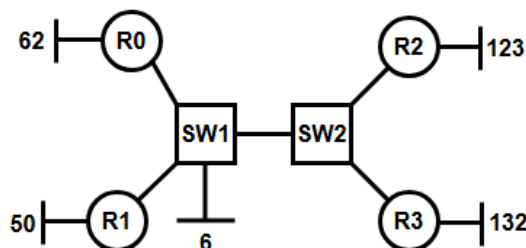
Answer: Subnet Mask: 255.128.0.0 (or /9).

II. Penny sends a packet to all devices on R1-LAN. Which destination IP shall she use? [2 Marks]

Solution: To reach all devices within her subnet, Penny uses the **directed broadcast address** of R1-LAN:

192.127.255.255

III. VLSM Subnetting for Expanded Topology with Root Network 69.0.0.0/15. [10 Marks]



Accounting for Router Interfaces:

- R3 LAN: 132 hosts + 1 gateway = **133 IP addresses**.
- R2 LAN: 123 hosts + 1 gateway = **124 IP addresses**.
- R0 LAN: 62 hosts + 1 gateway = **63 IP addresses**.
- R1 LAN: 50 hosts + 1 gateway = **51 IP addresses**.
- Switched Core (SW1–SW2): 6 hosts + 4 router interfaces (R0, R1, R2, R3) = **10 IP addresses**.

Allocation in Descending Order:

1. R3 LAN (Needs 133):

- $2^h - 2 \geq 133 \implies h = 8$ bits ($2^8 - 2 = 254$). Prefix: /24 (255.255.255.0).
- **Subnet Address:** 69.0.0.0/24 (Range: 69.0.0.1 – 69.0.0.254; Broadcast: 69.0.0.255)

2. R2 LAN (Needs 124):

- $2^h - 2 \geq 124 \implies h = 7$ bits ($2^7 - 2 = 126$). Prefix: /25 (255.255.255.128).
- **Subnet Address:** 69.0.1.0/25 (Range: 69.0.1.1 – 69.0.1.126; Broadcast: 69.0.1.127)

3. R0 LAN (Needs 63):

- $2^h - 2 \geq 63 \implies h = 7$ bits ($2^6 - 2 = 62 < 63 \implies h = 7$). Prefix: /25 (255.255.255.128).
- **Subnet Address:** 69.0.1.128/25 (Range: 69.0.1.129 – 69.0.1.254; Broadcast: 69.0.1.255)

4. R1 LAN (Needs 51):

- $2^h - 2 \geq 51 \implies h = 6$ bits ($2^6 - 2 = 62$). Prefix: /26 (255.255.255.192).
- **Subnet Address:** 69.0.2.0/26 (Range: 69.0.2.1 – 69.0.2.62; Broadcast: 69.0.2.63)

5. Switched Core Network (Needs 10):

- $2^h - 2 \geq 10 \implies h = 4$ bits ($2^4 - 2 = 14$). Prefix: /28 (255.255.255.240).
- **Subnet Address:** 69.0.2.64/28 (Range: 69.0.2.65 – 69.0.2.78; Broadcast: 69.0.2.79)

IV. For the switched network, how many addresses are we wasting? [2 Marks]

- Allocated /28 provides $2^4 - 2 = 14$ usable host addresses.
- Actually utilized: 6 hosts + 4 router interfaces = 10 addresses.
- **Wasted Usable Addresses:** $14 - 10 = 4$ usable addresses (or $16 - 10 = 6$ total block addresses including network and broadcast).

Question 2: IPv4 Fragmentation Header & Datagram Reconstruction [CO2, CO3]
4 + 3 + 2 = 9 Marks

Given Fragment Table:

#	Total Length	MF	DF	Offset
10	1544	1	1	1692
...	...	...	...	...
15	69	0	1	2632

I. Identify the header size of the fragment. [4 Marks]

Solution:

- From the course networking study notes, the Fragment Offset for the k -th fragment is given by:

$$\text{Fragment Offset}_k = \frac{(k - 1) \times \text{Payload}}{8}$$

where k is the fragment index ($k = 1, 2, 3, \dots$) and Payload is the data payload size per full fragment.

- For Fragment 10 ($k = 10$), the given fragment offset is $\text{Offset}_{10} = 1692$. Applying the formula:

$$1692 = \frac{(10 - 1) \times \text{Payload}}{8} = \frac{9 \times \text{Payload}}{8}$$

- Solving directly for Payload:

$$\text{Payload} = \frac{1692 \times 8}{9} = \frac{13,536}{9} = \mathbf{1504} \text{ bytes}$$

- (Cross-check with Fragment 15 ($k = 15$): $\text{Offset}_{15} = \frac{(15-1) \times 1504}{8} = \frac{14 \times 1504}{8} = 14 \times 188 = 2632$, matching the table exactly!)
- Given Total Length of Fragment 10 is 1544 bytes, the header size is:

$$\text{Header Size} = \text{Total Length} - \text{Payload} = 1544 - 1504 = \mathbf{40} \text{ bytes}$$

Answer: 40 bytes (20-byte base IPv4 header + 20-byte IP options).

II. Identify the packet size of the original datagram. [3 Marks]

Solution:

- Fragment 15 has Total Length = 69 bytes. With a 40-byte header, its data payload is:

$$\text{Data in Fragment 15} = 69 - 40 = 29 \text{ bytes}$$

- Total original payload data = $21,056 + 29 = 21,085$ bytes.
- Total size of the original datagram:

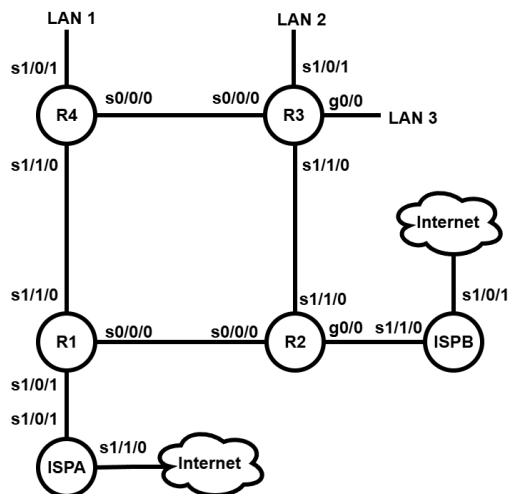
$$\text{Original Packet Size} = 21,085 + 40 = \mathbf{21,125} \text{ bytes}$$

Answer: 21,125 bytes.

III. Which two fields are required to orderly reassemble the fragmented packets? [2 Marks]

- Identification (ID) Field:** Associates all fragments belonging to the same datagram.
- Fragment Offset Field:** Determines the exact sequential position of each fragment's payload.

Question 3: Static Routing Configurations & Administrative Distance [CO2, CO3]
3 × 5 = 15 Marks



Device	Interface	IP	Network
R1	s0/0/0	.101	172.20.10.100/30
	s1/1/0	.225	172.20.11.224/30
	s1/0/1	.1	172.20.9.0/30
R2	s1/1/0	.193	172.20.12.192/30
	s0/0/0	.102	172.20.10.100/30
	g0/0	.1	178.51.100.0/30
R3	s1/0/1	.1	172.20.96.0/24
	g0/0	.1	172.20.80.0/26
	s1/1/0	.194	172.20.12.192/30
	s0/0/0	.97	172.20.10.96/30
R4	s1/0/1	.1	172.20.64.0/25
	s0/0/0	.98	172.20.10.96/30
	s1/1/0	.226	172.20.11.224/30
ISPA	s1/0/1	.2	172.20.9.0/30
	s1/1/0	.1	210.1.1.0/24
ISPB	s1/1/0	.2	178.51.100.0/30
	s1/0/1	.1	215.1.1.0/24

I. On R2, add a recursive static route to LAN1 via R1. [3 Marks]

- Destination: LAN1 = 172.20.64.0/25 (Subnet Mask: 255.255.255.128).
- Recursive static route specifies the next-hop IP on R1: 172.20.10.101.
- **Command:** R2(config)# ip route 172.20.64.0 255.255.255.128 172.20.10.101

II. Configure a backup route (directly attached) of the above static route (AD 6). [3 Marks]

- Backup path to LAN1 goes through R3 via interface s1/1/0.
- Directly attached specifies the exit interface with AD = 6.
- **Command:** R2(config)# ip route 172.20.64.0 255.255.255.128 s1/1/0 6

III. On R4, add a recursive default static route to go through the ISPB link. [3 Marks]

- Default route: 0.0.0.0 0.0.0.0. Next-hop IP on R3 is 172.20.10.97.
- **Command:** R4(config)# ip route 0.0.0.0 0.0.0.0 172.20.10.97

IV. If the above route wasn't configured on R4, what problem would have arisen? [3 Marks]

- R4 would have no route for unknown Internet destinations. Traffic from LAN1 destined for the Internet would have no match in R4's routing table, causing R4 to drop the packets and return ICMP Destination Unreachable errors.

V. Two static routes configured on R3: Which entry should be in R3's table and why? [3 Marks]

- **Selected Entry:** S 172.20.64.0/25 [20/0] via 172.20.12.193
- **Reason:** When multiple routes exist for the exact same destination prefix, the router selects the route with the **lowest Administrative Distance (AD)**. The route via 172.20.12.193 has an AD of 20, which is preferred over the route with AD 30.

SECTION B: ANY TWO OUT OF THREE (12 MARKS)

(All questions solved below for complete coverage)

Question 4: Distance Vector Convergence & Failure Advertisement [CO3] 2 + 4 = 6 Marks

I. Networks present in R4's routing table at cold start ($t = 0$): [2 Marks] Only its directly connected networks:

1. LAN1: 172.20.64.0/25 (connected to s1/0/1)
2. Link R4-R3: 172.20.10.96/30 (connected to s0/0/0)
3. Link R4-R1: 172.20.11.224/30 (connected to s1/1/0)

II. Failure of R3-R2 link and advertisement by R3: [4 Marks]

- **Information Shared:** R3 advertises that the route to ISPB via R2 is either unreachable (metric $\infty = 16$) or advertises its newly computed path through R4 with an updated metric.
- **Interfaces Advertised To:** R3 transmits this routing update across interface s0/0/0 towards Router R4.

Question 5: IPv6 Compression & All-Nodes Multicast [CO3] 1 + 1 + 1 + 3 = 6 Marks

I. Address Simplification:

- A. 2001:0db8:12af:0000:0000:0000:0a20:0004 → 2001:db8:12af::a20:4
- B. 0000:0000:0000:0000:0000:0000:0000:0001 → ::1
- C. Type of address for (B): **Loopback Address**.

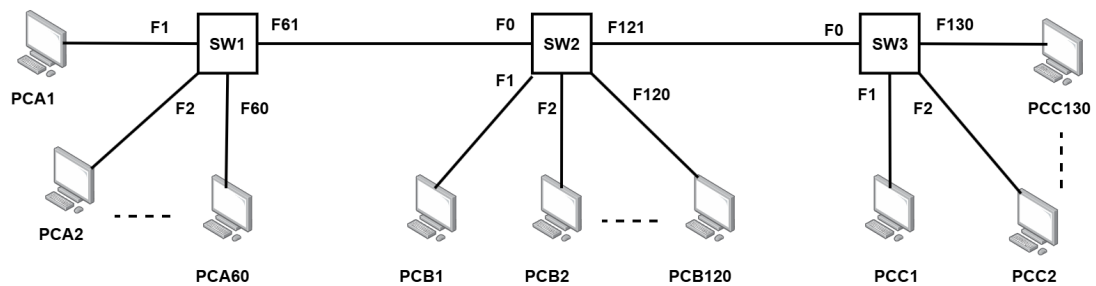
II. Sending to all devices on the network in IPv6 without broadcast: [3 Marks]

- IPv6 replaces traditional broadcast with the **All-Nodes Link-Local Multicast Address**:

FF02::1

- Every IPv6-enabled host on the local link automatically joins the FF02::1 multicast group. When a device needs to reach all nodes on the link (e.g., in Neighbor Discovery), it transmits the packet to destination FF02::1.

Question 6: Multi-Switch Broadcast Flooding & Switch Table Learning [CO2] 3 + 3 = 6 Marks



I. How many PCs will receive the ARP request frame? Why? [3 Marks]

- PCA1 broadcasts an ARP request for PCC2. Broadcast frames are flooded out of every active switch port across all connected switches (SW1, SW2, SW3).
- **PCs Receiving the Frame:**
 - On SW1: 60 PCs total – 1 (the sender PCA1) = 59 PCs.
 - On SW2: All 120 PCs (PCB1 to PCB120).
 - On SW3: All 130 PCs (PCC1 to PCC130).
- **Total PCs receiving the request:** $59 + 120 + 130 = 309$ PCs.

II. Switch Tables After PCA1 Receives the Reply: [3 Marks]

Switch SW1		Switch SW2		Switch SW3	
Device	Port	Device	Port	Device	Port
PCA1	F1	PCA1	F0	PCA1	F0
PCC2	F61	PCC2	F121	PCC2	F2

SECTION C: ANY THREE OUT OF FIVE (18 MARKS)

(All questions solved below for complete coverage)

Question 7: IPv4 Design Fundamentals

[CO2, CO3]

2 + 2 + 2 = 6 Marks

I. Why Fragment Offset is measured in multiples of 8 bytes: [2 Marks] The Fragment Offset field is only 13 bits wide. Since an IPv4 datagram can be up to $2^{16} - 1 = 65,535$ bytes long, scaling by 8 allows $2^{13} \times 8 = 65,536$ bytes to be indexed within a compact 13-bit header field.

II. Why IPv4 is unreliable: [2 Marks] IPv4 provides “best-effort” delivery without built-in acknowledgments, flow control, or retransmission mechanisms. If a packet encounters congestion or checksum corruption, it is dropped silently. Reliability is delegated to Layer 4 (TCP).

III. Scenarios where Traceroute helps more than PING: [2 Marks] PING only confirms whether a destination is reachable. Traceroute identifies the exact intermediate hop where a failure, excessive latency, or routing loop occurs.

Question 8: Centralized DHCP & DHCP Relay Agent

[CO2]

3 + 3 = 6 Marks

I. Why branch PCs cannot obtain IP addresses: [3 Marks] DHCPDISCOVER is a broadcast packet. Routers terminate broadcast domains and do not forward broadcast packets across WAN links to headquarters.

II. Role of the branch router in solving this issue: [3 Marks] Configure the branch router interface as a **DHCP Relay Agent** (`ip helper-address <HQ_DHCP_IP>`), which intercepts local broadcast discoveries, encapsulates them into unicast UDP packets, and forwards them across the WAN to the headquarters DHCP server.

Question 9: Carrier-Grade NAT (CGNAT) & Port Forwarding

[CO2]

3 + 3 = 6 Marks

I. Why friends cannot connect to your gaming server: [3 Marks] The public IP is shared among multiple customers via **Carrier-Grade NAT (CGNAT)**. The ISP router only tracks outbound sessions; unsolicited inbound connections from the Internet have no matching state entry in the ISP’s NAT table and are dropped.

II. Networking solution to allow direct reachability: [3 Marks] Acquire a **Dedicated Static Public IP address** from the ISP, or configure **Port Forwarding**, or use an encrypted reverse tunnel (e.g., Cloudflare Tunnel / VPN).

Question 10: IPv6 Encapsulation & Overhead

[CO2]

4 + 2 = 6 Marks

I. IPv6 Tunneling across IPv4: [4 Marks] IPv6 packets are encapsulated inside IPv4 packets at the edge router, allowing them to traverse IPv4-only core networks without modifying intermediate routers. This enables cost-effective, incremental transition to IPv6.

II. Problems with repeated encapsulation/decapsulation: [2 Marks] Adds header overhead (reducing effective payload MTU), increases risk of packet fragmentation, and adds CPU processing delay at tunnel endpoints.

Question 11: Switch MAC Aging Timer & Port Migration [CO2] 3 + 3 = 6 Marks

I. Why the PC couldn't print immediately: [3 Marks] The switch MAC address table cached the printer's MAC address on the old port. Until the ****MAC Aging Timer**** (typically 300 seconds / 5 minutes) expired, the switch continued forwarding print frames to the old port.

II. What could have been done to enable immediate printing: [3 Marks] Manually clear the switch MAC table (`clear mac address-table dynamic`), or trigger an immediate frame transmission (such as a Gratuitous ARP or ping) from the printer so the switch instantly learns the new port.

— END OF EXAMINATION SOLUTION —